Band-1 Hadamard Order 9872 – Classical Paley-I (q=9871); IN SPK via Paley-I; Gap 2052 Still OPEN

H_9872 Paley-I PASS; Paley-II FAIL/SKIP (q=4935=3*5*7*47 *NOT prime / not prime power; ===3 mod 4); Kronecker ELIGIBLE/SKIP construction (4*2468; 2468 in ONBOX_SEEDS + admissible; Paley-I claim path); Miyamoto digression skipped; gap 2052 OPEN; conjecture OPEN; GATE-04 LOCKED

C.M. Baird; ZoraASI mission-control Band-1 dash

2026-09-23

Document class. Scientific computational / construction certificate (Band-1).
Program. Open-mathematics collapse list * Human+AI attack order * ZoraASI mission-control.
Parents. `2026-09-23_Band1_Hadamard_9860_PaleyI`; `2026-09-22_Band1_Hadamard_9852_PaleyI`; `2026-09-22_Band1_Hadamard_9840_PaleyI`.
Lab. `/workspace/experiments/band1/hadamard_9872_paleyI/`.
Cutoff. 2026-09-23T07:10:12Z (~3:10AM ET).

Honesty stamp (non-negotiable).
This memo claims **finite construction of order 9872 only (verified max|HH^T-nI|=0), via classical Paley-I q=9871 (prime ===3 mod 4). 9872 IN SPK via Paley-I.** Honest skip: Miyamoto digression this wave (prefer classical SPK). Classical SPK for 9872: Paley-I **PASS** ($q = 9871$ prime $\equiv 3 \pmod 4$); Paley-II **FAIL/SKIP** ($q = 4935 = 3 \cdot 5 \cdot 7 \cdot 47$ NOT prime / not prime power; $\equiv 3 \pmod 4$); Kronecker **ELIGIBLE/SKIP construction** (note $9872 = 4 \cdot 2468$; $2468 \equiv 0 \pmod 4$ admissible AND in ONBOX_SEEDS; Paley-I claim path). **9872** folds into SPK **only** via classical Paley-I (this run). It **does not** claim the full Hadamard conjecture. GATE-04 collection remains **LOCKED**. No RH / P≠NP / YM / Frankl / NS-solved / AC-solved claims. Gap **2052** remains **OPEN**. `clay_closed = false`. Prior SPK **9860+9852+9840+9812+9804+9792+9788+9768+9744+9740+9720+9680+9644+9632+9624+9620-** retained; Miyamoto-only **8948/8644/8516/8452/7988/...** excluded.

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Abstract

Finite construction of order **9872**.

- H_{9872} = Paley-I ($q = 9871$) – classical SPK

- Paley-II for 9872: **FAIL/SKIP** ($q = 4935 = 3 \cdot 5 \cdot 7 \cdot 47$ NOT prime / not prime power; $\equiv 3 \pmod 4$)
- Kronecker for 9872: **ELIGIBLE/SKIP construction** ($4 \cdot 2468$; 2468 in ONBOX_SEEDS + admissible; Paley-I claim path)
- Skip: Miyamoto digression this wave

SPK: IN SPK via Paley-I.
Gap 2052: OPEN. **Conjecture:** OPEN. **clay_closed:** false. **GATE-04:** LOCKED.

```
python3 verify_hadamard_9872.py
```

Results. overall_pass=True; units 171/171; row blocks 78/78 (block size 128); wall 781.5135s; constructed max|HHˆT-nI|=0.

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Arithmetic

- Target $n = 9872 = q + 1$ with $q = 9871$ prime $\equiv 3 \pmod 4$ — Paley-I eligible.
- Paley-II: $q = 4935 = 3 \cdot 5 \cdot 7 \cdot 47$ NOT prime / not prime power; $\equiv 3 \pmod 4$ — FAIL/SKIP.
- Kronecker: $9872 = 4 \cdot 2468$; $2468 \equiv 0 \pmod 4$ admissible AND in ONBOX_SEEDS — ELIGIBLE/SKIP construction (Paley-I claim path).
- Miyamoto digression skipped this wave (prefer classical SPK).

Constructions (verified)

Object	Method	max|HHˆT-nI|	Wall (s)
H_9872 Paley-I	Paley-I q=9871	0	764.1962
H_9872 Paley-II	FAIL/SKIP (q=4935=3*5*7*47 NOT prime / not prime power; ===3 mod 4)	–	–
H_9872 Kronecker	ELIGIBLE/SKIP construction (4*2468; 2468 in ONBOX_SEEDS + admissible; Paley-I claim path)	–	–

SPK closure note

Seeds include prior SPK through **9860** lineage (**9860**; **9852**; **9840**; **9812**; **9804**; **9792**; **9788**; **9768**; **9744**; **9740**; **9720**; **9680**; **9644**; **9632**; **9624**; **9620**; **9588**; **9552**; **9548**; **9540**; **9512**; **9492**; **9480**; **9468**; **9420**; **9404**; **9392**; **9372**; **9344**; **9324**; **9320**; **9312**; **9284**; **9240**; **9228**; **9204**; **9200**; **9188**; **9152**; **9104**; **9128**; **9124**; **9092**; **2220+8884**; **2236**; **2160**; **2140+8564**; ...; **NOT** 8948/8644/8516/...) + verified **9872** (+ prior **9860+9852+9840+9812+9804+9792+9788+9768+9744**). Next constructive gap **2052** (OPEN).

Quantity	Value
9872 in SPK	**True** (Paley-I q=9871)
Gap 2052	**OPEN**
clay_closed	false
GATE-04	LOCKED

Honesty

- clay_closed = false
- GATE-04 LOCKED
- Hadamard conjecture OPEN
- Gap 2052 OPEN (method triage / non-finding only)
- No RH / P!=NP / YM / Frankl / NS / AC solved claims
- SPK membership only via classical Paley-I/II/Kronecker (not Miyamoto alone)

Next optional

q=9883->n=9884 Paley-I DIRECT q=9883 OR q=9887->n=9888 Paley-I DIRECT q=9887 OR n=2052 standard SPK constructive gap – method-attack / lit OR q=9907->n=9908 Paley-I DIRECT q=9907 OR Track B / Object 1 meter await